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CSE420

Assignment 2

Section 01

Ans no 1

Input : $\text{int}[5]a; \rightarrow \text{int}[\text{num}]id;$

Stack	Symbol	Input	Action
0	\$	$\text{int}[\text{num}]id; \$$	shift to 4
04	\$ int	$[\text{num}]id; \$$	reduce by $B \rightarrow \text{int}$
03	\$ B	$[\text{num}]id; \$$	shift to 8
038	\$ B [	$\text{num}]id; \$$	shift to 10
038 10	\$ B [num	$]id; \$$	shift to 12
038 10 12	\$ B [num]	$id; \$$	reduce by $e \rightarrow \epsilon$
038 10 12 13	\$ B [num] C	$id; \$$	reduce by $C \rightarrow [\text{num}]C$
037	\$ B C	$id; \$$	reduce by $T \rightarrow BC$
02	\$ T	$id; \$$	shift to 6
026	\$ T id	$i \$$	shift to 9
0269	\$ T id;	$\$$	reduce by $D \rightarrow \epsilon$
0269 11	\$ T id; D	$\$$	reduce by $D \rightarrow T id; D$
01	\$ D	$\$$	Accept

The parser reaches the Accept state.

Therefore, $\text{int}[5]a;$ is a valid input string according to the grammar.